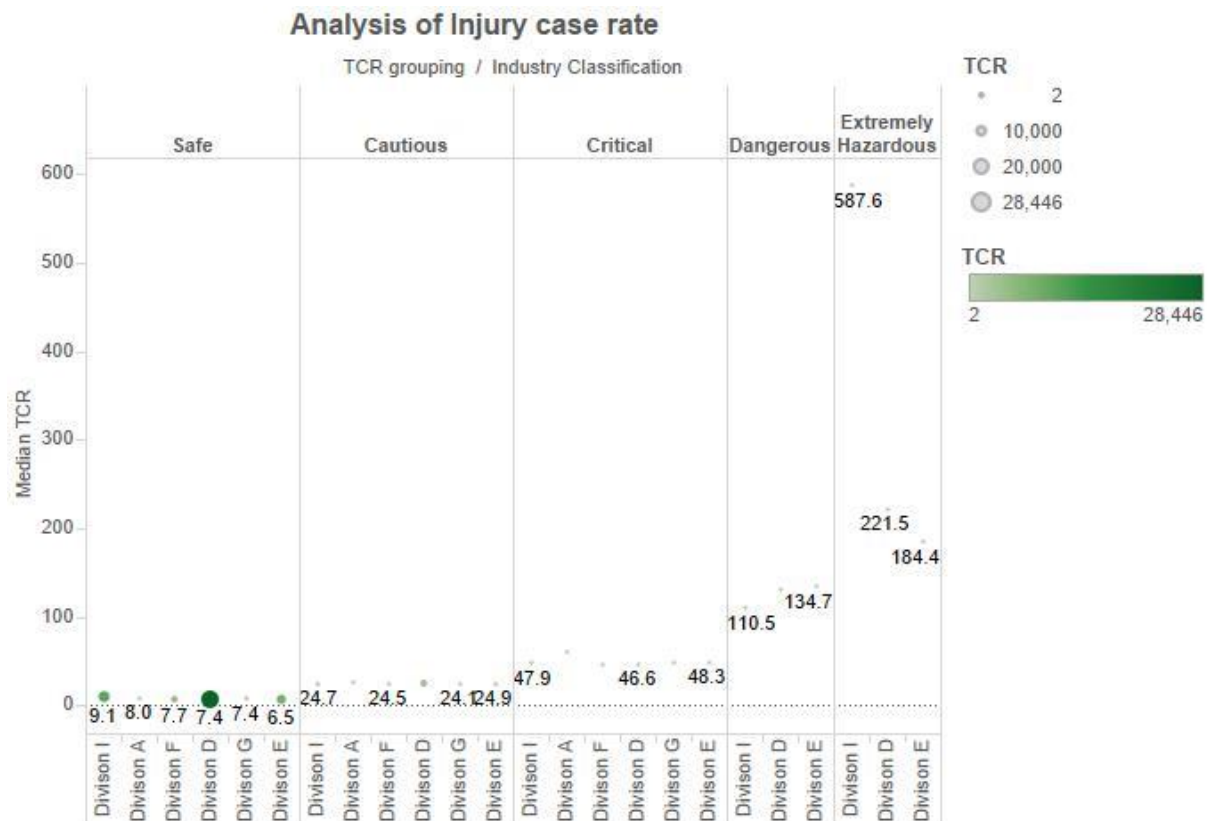


MOOC FINAL ASSIGNMENT- STRATEGIC BUSINESS ANALYTIC

The data is taken from OSHA database and is from the year 2002. It gives the detail of companies along with work related injuries rate such as Total case rate (TCR), Days away restricted and transfer (DART) and Days away from work (DAFWII). The SIC number in the data was used to identify the division of industry as per the OSHA site.



Median of TCR for each Industry Classification broken down by TCR grouping. Color shows count of TCR. Size shows count of TCR.

Looking at the frequency distribution of TCR from the data it was striking to note that maximum count of TCR fell in the range of 7 to 50 (mean TCR was 10.7). In this graph the injuries (TCR) occurring in a company from different division (SIC - classified as per OSHA site) is grouped into five categories and plotted against the median of TCR. **Almost 87% of the companies lies in the safe group with a median TCR of 7.7.**

Groups	TCR
Safe	0~19
Cautious	20~49
Critical	50~99
Dangerous	100~149
Extremely hazardous	>150

The injury occurring at the company cost the employer in terms of loss of productivity and compensation paid to the employee while the employee loses his/her opportunity to earn a salary. To protect the employee from this burden and risk, work insurance is an absolute requirement.

The cost of these injury depends upon the type and a serious injury can result in days away from work which is reflected through DAFWII. Since there is an underlying cost to these injury it will be interesting to identify the companies falling in the dangerous and extremely hazardous groups. As the value of TCR increases the company possibility to pay higher compensation to the employees also increases. Thus **Insurance Company can charge higher premium from companies such as in the dangerous group (Division D, E & I)** and hence can use this data to assess the company safety status. By categorising the companies within groups (mentioned above) and developing an algorithm it can efficiently increase or decrease the premium without much efforts.

By Arun Joshi